Exploring Generative AI and ChatGPT

Introduction to Generative AI

- Generative AI refers to artificial intelligence models that can create new content: text, images, music, or videos based on the data they've been trained on.
- Examples
  - ChatGPT → generates human-like text.
  - DALL·E / Midjourney → generates images.
  - MusicLM → composes music.
  - Synthesia → creates AI videos.

Where do you think you've already seen Generative AI being used?

Introduction to ChatGPT

- ChatGPT is a Large Language Model (LLM) developed by OpenAI, trained to understand and generate natural language text.

- How It Works
  - Input: You type a message (called a *prompt*).
  - Processing: The AI predicts the most likely next words using billions of parameters.
  - Output: It gives you a clear and context-aware response.

- Uses of ChatGPT
  - Homework assistance and explanations
  - Writing summaries, essays, and reports
  - Brainstorming project ideas
  - Coding and debugging
  - Translating and paraphrasing

What is Prompt Engineering?

- It's about knowing *how* to talk to the AI to unleash its full potential.
- A good prompt is clear, specific, and provides context to reduce ambiguity.

Type of Prompt	Description	Example
Zero-Shot Prompting	You ask the question directly, assuming the AI already knows the task.	“Translate the following English sentence to French: 'The sun is shining.'”
Few-Shot Prompting	You provide a few examples of the desired input/output format within the prompt.	Example 1: Input: 'Happy'; Output: 'Positive'. Example 2: Input: 'Sad'; Output: 'Negative'.
Chain-of-Thought (CoT)	You instruct the model to show its step-by-step reasoning before giving the final answer. This greatly improves accuracy on complex tasks.	“Calculate 53 x 12. Show your work step-by-step before the final answer.”
Role-Playing Prompt	You assign the AI a specific persona or role to influence the tone and style of the output.	“Act as a leading Silicon Valley Venture Capitalist. Write a pitch for a new AI-powered coffee machine.”

Fun Examples

- Write a rap song about mid-semester exams.
- Imagine AI and humans are best friends: write their WhatsApp chat.
- Explain ICT as if you’re a pizza chef.
- Write funny urdu poetry on AICT course

BONUS (using account)
Roast me based on my previous prompts, no mercy, no filter, straight to my soul جواب اردو میں دینا

Activity 1: Create an AI Image

- Objective: Understand how text prompts can generate visuals.
- Steps:

1. Open ChatGPT or Gemini.
2. Give a base prompt: “Create an image of a futuristic classroom where humans and robots learn together.”
3. Compare results from ChatGPT and Gemini image generators.
4. See how changing words (e.g., “cartoon,” “realistic,” “cyberpunk”) changes the result.

Activity 2: Generate a Presentation Prompt

- Objective: Learn to use AI for productivity.
- Steps:

1. Ask ChatGPT: “Create a prompt for a presentation on the impact of AI on education.”
2. Take that generated prompt and paste it into PPT.ai (or SlidesGPT, Gamma.app, etc.).
3. See how the AI auto-generates slides and structure.

Suggested Homework

- Write one creative prompt and one academic prompt.
- Try both in ChatGPT and Gemini, then compare which one gives a more relevant output.